

Homework 8

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GitHub repo link (CLICK)

Problem 1. regression warm up

Part A

When we use the `lm()` function calculator in R, a linear regression model, we get the output of 147.81 as the intercept and -0.61 as the rate of creatine clearance in patients depending on age. This gives us the equation $creatclear = -0.62(age) + 147.81$. Using this equation, when a patient's age is 55, their expected creatine clearance rate is about 113.72 mL/min.

Part B

When we use the `lm()` function in R, the summary displaces the rate— which we used in Part A. The creatine clearance rate is -0.62 mL/min per year, which means that a patient's creatine clearance rate decreases by about 0.62 mL/min every year.

Part C

Using the equation from Part A: $creatclear = -0.62(age) + 147.81$

40 year-old patient calculations:

$$creatclear = -0.62(age) + 147.81$$

$$135 = -0.62(40) + 147.81$$

$$135 > 123.01$$

$$\text{Difference} = 11.99$$

60 year-old patient calculations:

$$creatclear = -0.62(age) + 147.81$$

$$112 = -0.62(60) + 147.81$$

$$112 > 110.61$$

$$\text{Difference} = 1.39$$

The 40 year-old is healthier for their age because their creatine clearance is 11.99 higher than expected, compared to the 1.39 difference from the 60 year-old patient.

Problem 2. Modeling disease growth

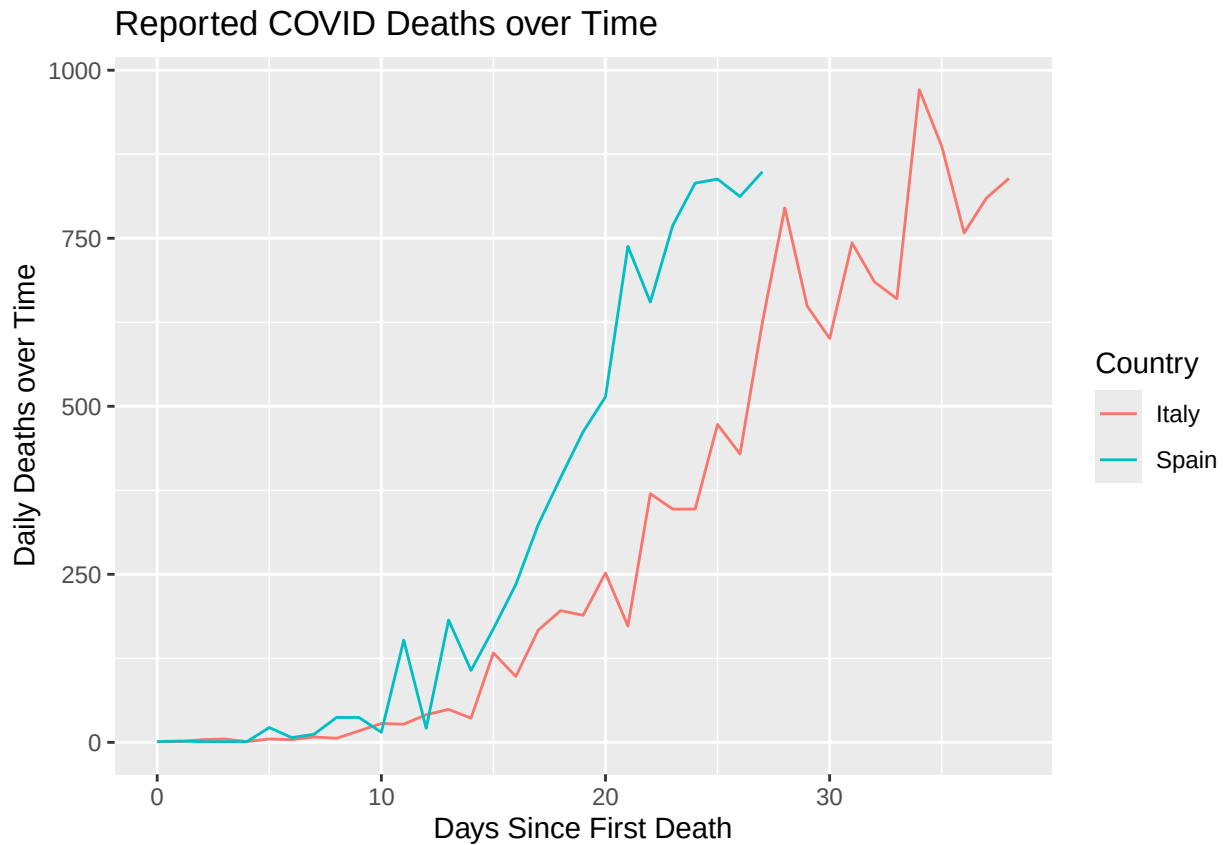
Part 1

For Italy, the estimated growth rate for deaths caused by COVID is 0.183 and the doubling time is 3.8 days.

Part 2

For Spain, the estimated growth rate for deaths caused by COVID is 0.276 and the doubling time is 2.5 days.

Part 3



Problem 3. price elasticity of demand

The estimated price of elasticity of demand for milk is -1.62.